Martin McEnroe, P.E.

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TECHNOLOGY EXECUTIVE—ARTIFICIAL INTELLIGENCE

Two U.S. Patents Filed 2026—AI Agent Safety • 22 Patents Issued • Licensed P.E. • IEEE Co-Chair, AI for the Power Grid

Technology executive who writes industry standards for artificial intelligence and builds production systems. Co-Chairs an emerging technologies subcommittee of the Long Range Planning Committee at the IEEE Power & Energy Society, the world's largest power engineering body. Directed data science and artificial intelligence at AT&T across a 45-person organization, a $9M annual budget, and $10M in recurring annual savings. Holds 22 issued U.S. patents and filed two in 2026 on AI agent safety, ahead of the standards. Designs and ships production AI products through the Chrome Web Store, Mozilla Add-ons, and the Python Package Index. Licensed Professional Engineer.

IEEE LEADERSHIP AND STANDARDS

Co-Chair, Emerging Technologies, Digitalization, and AI Readiness—IEEE Power & Energy Society, Long Range Planning Committee 2026–present

Forecast the technologies reaching the power system over the next five to fifteen years, and advise the Society on where to act. Convene fourteen voting members drawn from industry, universities, and government across eight countries. Delivered the subcommittee's Vision Document, its agenda for AI and digitalization in the power system: members reviewed candidate priorities and voted independently on each, adopting 234. Twelve areas now under investigation, among them trustworthy and certified AI, cyber-physical security, quantum computation, and autonomous closed-loop grid operation.

Officer and Lead Author, Data Centers and the Grid—IEEE Standards Association 2025–present

Activity Secretary of the IEEE Standards Association program on data center growth, the sector's most urgent capacity problem. Vice Chair of the first white paper, published January 2026 (ieeexplore.ieee.org/document/11366058). It triggered an IEEE study group that approved five new standards projects. One of four lead authors on the second paper, now in editorial review.

Standards Author and Working Group Member—IEEE Power & Energy Society 2025–present

Serve on four working groups across the security and data agenda. Authored the use-case section of the report on software bills of materials in the energy sector, grounded in federal guidance from the Cybersecurity and Infrastructure Security Agency, the National Security Agency, and NERC. Authored a section of the joint task force report naming which new communications standards should exist. Founding member of IEEE P4200, which sets the interconnection, performance, and interoperability requirements for attaching data centers to transmission and distribution systems. Contributing author on the large-load integration report.

THRIVETECH.AI—APPLIED AI PORTFOLIO

Multi-Agent Engineering Systems—AssemblyZero

- Built an engineering organization staffed by AI agents. AssemblyZero builds every product in this portfolio, and built and improves itself. Five stages run from triage through design, specification, test-driven development, and pull request, coordinating twelve or more concurrent agents with typed state, conditional routing, and crash recovery.
- Designed cross-vendor adversarial verification, the core idea: one vendor's model writes the code, a different vendor's model reviews it. Errors that survive one model rarely survive two trained by separate companies.
- Built five evaluation layers that every change must clear: test execution, structural analysis of the code, cross-model review, stagnation detection, and learning from accumulated verdicts.
- Processed 1,578 issues at a 91% closure rate, merged 1,031 pull requests, and grew the test suite to 9,982. Python, LangGraph, DynamoDB, GitHub Actions.

Adaptive Learning Systems—Chiron (chiron.thrivetech.ai)

- Built a live learning platform running eleven subject domains, from AI governance and patent law to real analysis, music theory, and amateur radio licensure.
- Designed a question generator that works from cited source material, classifies each item against Bloom's Revised Taxonomy and Webb's Depth of Knowledge, and tags every wrong answer with the specific misconception it represents under the SOLO taxonomy.

- Designed a knowledge model spanning all eleven domains instead of siloing them: a shared ontology, a prerequisite graph, and per-learner mastery state.
- Ensured no learner meets a question resting on a concept they have not established. Introduces and teaches new concepts at the learner's knowledge frontier. Ranked selection by unseen material, then by decayed per-concept mastery.
- Wrote the research behind it, a cross-domain study of which cognitive frameworks survive automated question generation, including where Bloom's levels alone let a model produce an item that only looks analytical.

Language and Culture—thetrail.online

- Built a subscription product that teaches language and culture along real long-distance walks, opening with the Camino Francés in Spanish. A walker moves stage by stage; encounters recur; the vocabulary is the vocabulary of hospitality.
- Designed the motivation to come from the walker rather than from streaks and mascots. People already dream of a specific walk, and every walk carries a language.
- Structured each stage across four levels, gated on proficiency, with click-to-hear audio on every word.
- Built a human curation gate: no image, recording, or encounter reaches a learner without review.

AI for Critical Infrastructure—Palaestra and Exedra (palaestra.thrivetech.ai)

- Built Palaestra to measure how best large data centers can flex their demand under grid stress, the open question in the IEEE data center work above.
- Designed a competition in which a player runs a data center at the grid interface for a simulated day, scored against a certified optimal line. Human or AI agent; both speak the same interface. Built five scenarios on public reliability testbeds.
- Published the scenario format, the competitor interface, and the complete engine rules as Exedra (github.com/martymcenroe/Exedra), so any group can build a scenario from its own region's public grid records. Wrote a multi-lesson curriculum.

Consumer and Developer Products—Shipped Through Three Public Registries

- Shipped **Aletheia** to both the Chrome Web Store and Mozilla Add-ons: select any text on any page and get the etymology and historical context concealed inside it. Named for Heidegger's term for unconcealment.
- Designed a model ladder the reader climbs on request. A cheap model makes the first pass. Each request for more depth moves the work up a tier, and the analysis unconceals at ever-greater depths of meaning and poetry.
- Screened every request through a defense funnel before any analysis runs: selection check, hashed denylist, then a semantic classifier. Tested against eight attack classes. Limited logging to operational metadata, never content.
- Built **Clio**, a Chrome Web Store extension that extracts conversations with language models into structured data.
- Published **Silphe** to the Python Package Index, a behavioral biometrics library that quantifies a visuomotor signature from pointer dynamics: acquisition under Fitts's law, hold tremor, and tracking lag. Kept analysis on the device and shipped with no third-party runtime dependencies.

Agent Safety Research

- Filed two U.S. patents in 2026 on AI agent safety, as independent inventor and applicant.
- Mapped AI agent failure modes onto MITRE ATT&CK insider-threat techniques. An agent pushed to finish a task behaves like an insider. Built a 42-probe audit tool across eight tactics and hardened a 50-repository fleet against what it found.
- Designed **sentinel-rfc**, an architecture giving agents their own permissions rather than inheriting the user's. Built **Maieutics**, a method for proving human authorship of AI-assisted technical work, anchored in USPTO inventorship guidance.

PROFESSIONAL EXPERIENCE

THRIVETECH.AI – Plano, TX June 2022–present

Founder & President

Independent AI architecture practice and consulting vehicle; holds the invention portfolio. Products and research are listed above under Applied AI Portfolio.

TEXAS DEPARTMENT OF TRANSPORTATION – Austin, TX September 2023–April 2024

AI Strategic Consultant

- Authored 76-page AI strategic plan encompassing 32 epics, 213 use cases, governance controls, risk assessments, and AI ethics framework for state-wide transportation agency.

- Conducted stakeholder interviews across every division and district; applied LDA topic modeling to surface themes and identify high-priority AI applications meeting federal and state regulatory requirements.
- Translated technical AI capabilities into business-value propositions for non-technical executives; established phased implementation roadmaps balancing feasibility, compliance, and organizational change.

AFINITI – Washington, DC March 2021–June 2022

Vice President, Product

Product and security leadership for AI-powered customer-agent pairing platform optimizing enterprise contact center operations through behavioral analytics and machine learning.

- Led security architecture initiatives on a ~1M-line, 10-year-old C/C++ codebase: SonarQube vulnerability remediation, penetration test remediation, architecture reviews, API security hardening.
- Defined cross-functional product development processes spanning engineering, AI research, MLOps, and go-to-market. Hands-on DevOps across GCP and Azure (Docker, Jira, CI/CD).
- Set product strategy and prioritization across quarterly and sprint planning cycles, balancing ML algorithm investment against operational improvements and production issues.

AT&T – Plano, TX September 1991–October 2020

Director, Data Science & Artificial Intelligence September 2017–October 2020

Hands-on architect and individual contributor leading enterprise-scale AI/ML solutions. Led 45-person cross-functional team, $9M annual labor budget, $10M+ recurring annual savings, $20M+ in initiative funding secured through SVP-level partnerships.

- Built **TAVI**, an NLP customer-service chatbot handling 10,000+ daily interactions at 92% accuracy. Cut costs $4.5M in 15 months and improved first-call resolution 30%. Python, TensorFlow, BERT, AWS Lambda, DynamoDB.
- Delivered **SNAP**, computer vision for field operations and the first cloud-native AI solution at AT&T enterprise scale. Saved $5.8M annually and raised field productivity 30%. Python, Azure Computer Vision, Core ML.
- Architected **EasyBake**, a no-code chatbot platform on Azure with API and robotic-process-automation overlays into dozens of enterprise systems. Won three internal production customers.
- Designed **Label Quest**, a gamified human-in-the-loop labeling system adopted by dozens of internal data science teams, covering text, image, and video with live leaderboards, accuracy tracking, and consensus quality gates.
- Delivered 90+ proof-of-concept solutions over three years; security protocols, audits, and vulnerability remediation for all AI deployments.

Lead Member Technical Staff, Enterprise Architect May 2013–August 2017

- Founded and chaired the enterprise AI & Data Science Architecture Review Board, establishing the governance gates AI systems passed through across business units of a Fortune 10 company. Evaluated frameworks, vendors, and architectures for security, interoperability, licensing, and cost. Drove Python and R to enterprise-approved status.
- Deployed and administered Stack Overflow Enterprise to 31,000 users, clearing Technical Architecture Board approval, security review, and privacy compliance.
- Managed ‘RCloud’, the internal data science platform, and led training on Python, R, APIs, and AI architectures.
- Led tech evaluations setting enterprise direction: R vs SAS, MicroStrategy vs Tableau, GitHub vs Atlassian, AWS vs Azure vs GCP.

Additional Roles (1991–2013)

AT&T Bell Labs R&D (Holmdel, NJ; 1991–1995) • Director of Program Management, IP Backbone • Director of Emerging Video Services, Corporate Strategy • Network Operations & Engineering • Wireless communications development (multiple patents) • Resolved year-long migration crisis with 100-fold efficiency improvement in 45 days • Secured SVP funding for AI initiatives.

LICENSURE

Licensed Professional Engineer—State of Texas

CERTIFICATIONS

AWS—7 Certifications including:

- Machine Learning–Specialty
- Solutions Architect–Professional
- Data Analytics–Specialty
- Data Engineer Associate

Microsoft Azure:

- DevOps Engineer Expert
- AI Fundamentals
- Security, Compliance & Identity Fundamentals

CISSP & **CCSP**—ISC2 | **PMP** & **PgMP**—PMI | **Wireless Communications Professional (WCP)**

EDUCATION

Master of Computer Science, University of Illinois Urbana-Champaign (2017)

Master of Science, Electrical Engineering, University of Maryland, College Park

Bachelor of Science, Electrical Engineering, University of Maryland, College Park

PROFESSIONAL MEMBERSHIPS

IEEE Senior Member • International Association of Privacy Professionals (IAPP) • ISC2 • Project Management Institute (PMI) • Open Worldwide Application Security Project (OWASP)

PATENTS

22 issued U.S. patents spanning AI, cloud computing, cybersecurity, IoT, and wireless communications, plus 9 pending applications and the two 2026 agent-safety filings. Domains include document analysis using ML and neural networks, customer service automation, sentiment detection, autonomous vehicle systems, edge network management, and biometric systems.